

**Report on Activity 12: Logistic Regression on Supsup et al. (2021) paper entitled
“Geographic distribution, population and conservation status of *Tragulus nigricans*, an
endangered mouse deer endemic to western Philippines”**

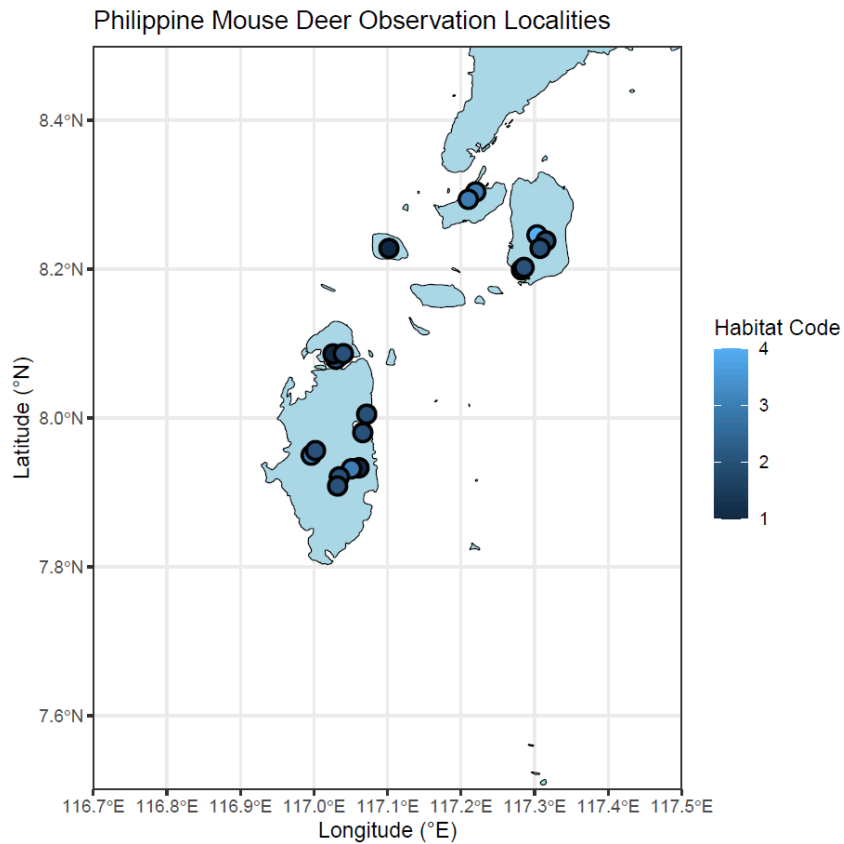


Figure 1. Geographic Distribution of Philippine Mouse Deer Observations in Palawan

Multiple Logistic Regression Curve: Average Distance

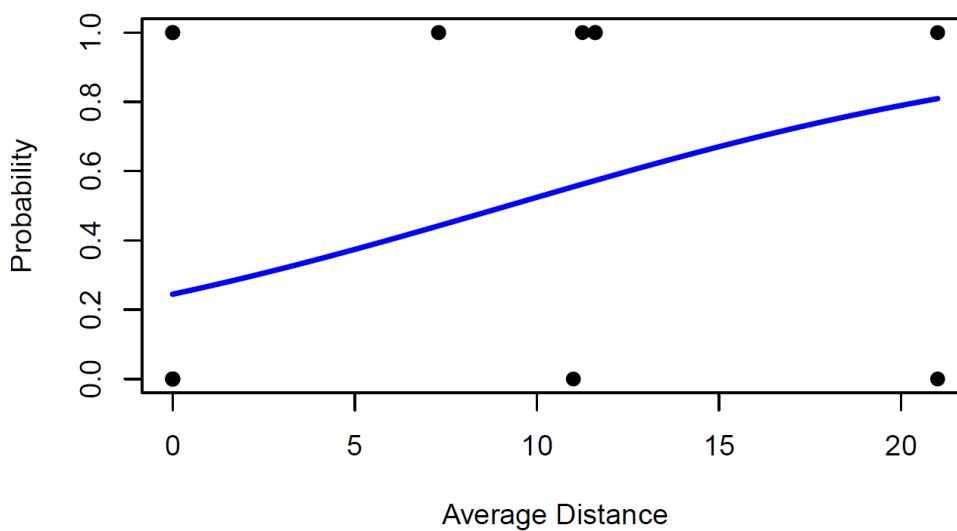


Figure 2. Multiple Logistic Regression Curve for Average Distance

Multiple Logistic Regression Curve: Sampling Effort

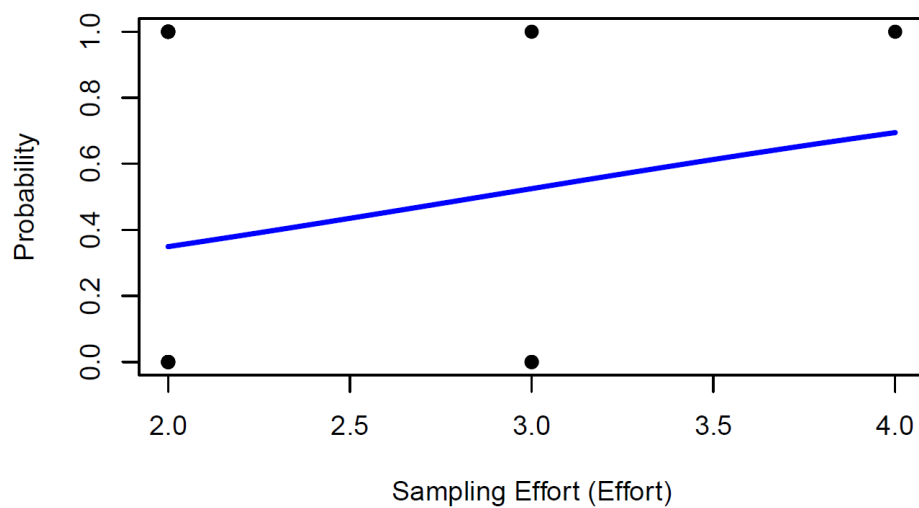


Figure 3. Multiple Logistic Regression Curve for Sampling Effort

Interpretation

Based on the generated graphs, the detectability of the Philippine Mouse Deer is influenced by the observation distance and sampling effort. When examining the data for the average distance, there is an inverse relationship between observation distance and species observation; wherein, the higher the distance of observation, the lower the chance of species observation. In terms of the sampling effort data, there is a direct relationship between sampling effort and the probability of species detection, which is seen when higher sampling effort also has a higher species detection rate. Therefore, increased sampling effort and lower observation distance can be inferred to increase species detection.